

--- CREATE DATABASE

=====

```
CREATE DATABASE Techno1;
USE Techno1;
```

/*=====

CREATE DEPARTMENT TABLE

=====*/

```
CREATE TABLE Department
```

```
(
    DeptID INT PRIMARY KEY,      -- Unique Department ID
    DeptName VARCHAR(50) NOT NULL -- Department Name
);
```

/*=====

CREATE EMPLOYEE TABLE

=====*/

```
CREATE TABLE Employee
```

```
(
    EmpID INT PRIMARY KEY,      -- Unique Employee ID
    EmpName VARCHAR(50) NOT NULL, -- Employee Name
    Salary DECIMAL(10,2),      -- Employee Salary
    DeptID INT,                -- Department ID (Foreign Key)
    HireDate DATE,             -- Employee Joining Date

    FOREIGN KEY (DeptID)
    REFERENCES Department(DeptID)
);
```

/*=====

INSERT DATA INTO DEPARTMENT TABLE

=====*/

```
INSERT INTO Department (DeptID, DeptName)
```

```
VALUES
```

```
(1, 'HR'),
```

```
(2, 'IT'),
```

```
(3, 'Finance'),
```

```
(4, 'Sales'),
```

```
(5, 'Marketing'),
```

```
(6, 'Admin'),  
(7, 'Operations'),  
(8, 'Support'),  
(9, 'Legal'),  
(10, 'Research');
```

```
/*=====
```

```
INSERT DATA INTO EMPLOYEE TABLE
```

```
=====*/
```

```
INSERT INTO Employee (EmpID, EmpName, Salary, DeptID, HireDate)  
VALUES  
(101, 'Megha', 75000, 2, '2026-02-10'),  
(102, 'Rahul', 65000, 1, '2025-12-15'),  
(103, 'Amit', 90000, 2, '2026-01-20'),  
(104, 'Priya', 85000, 3, '2025-11-10'),  
(105, 'Sneha', 60000, 4, '2026-03-05'),  
(106, 'Vikas', 95000, 2, '2025-10-01'),  
(107, 'Neha', 70000, 5, '2026-04-12'),  
(108, 'Kiran', 55000, 1, '2026-05-01'),  
(109, 'Rohan', 80000, 3, '2026-01-15'),  
(110, 'Pooja', 68000, 4, '2026-06-01');
```

```
/*=====
```

```
DISPLAY ALL RECORDS
```

```
=====*/
```

```
-- View Department Table  
SELECT *  
FROM Department;
```

```
-- View Employee Table  
SELECT *  
FROM Employee;
```

Results		Messages	
	DeptID	DeptName	
1	1	HR	
2	2	IT	
3	3	Finance	
4	4	Sales	
5	5	Marketing	
6	6	Admin	
7	7	Operations	
8	8	Support	
9	9	Legal	
10	10	Research	

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01

```

/*=====
1. TOP 5 HIGHEST SALARY EMPLOYEES
=====*/

```

```

SELECT TOP 5 *
FROM Employee
ORDER BY Salary DESC;

```

	EmpID	EmpName	Salary	DeptID	HireDate
1	106	Vikas	95000.00	2	2025-10-01
2	103	Amit	90000.00	2	2026-01-20
3	104	Priya	85000.00	3	2025-11-10
4	109	Rohan	80000.00	3	2026-01-15
5	101	Megha	75000.00	2	2026-02-10

```

/*=====
2. DEPARTMENT-WISE EMPLOYEE COUNT
=====*/

```

```

SELECT
    d.DeptName,
    COUNT(e.EmpID) AS EmployeeCount
FROM Department d
INNER JOIN Employee e
    ON d.DeptID = e.DeptID
GROUP BY d.DeptName;

```

	Results	Messages
	DeptName	EmployeeCount
1	Finance	2
2	HR	2
3	IT	3
4	Marketing	1
5	Sales	2

```

/*=====
3. SECOND HIGHEST SALARY
=====*/

```

```

SELECT MAX(Salary) AS SecondHighestSalary
FROM Employee
WHERE Salary <
(
    SELECT MAX(Salary)
    FROM Employee
);

```

	SecondHighestSalary
1	90000.00

```

/*=====
4. EMPLOYEES WHOSE SALARY IS GREATER
   THAN THEIR DEPARTMENT AVERAGE
=====*/

```

```

SELECT *
FROM Employee e

```

```

WHERE Salary >
(
    SELECT AVG(Salary)
    FROM Employee
    WHERE DeptID = e.DeptID
);

```

Results		Messages			
	EmpID	EmpName	Salary	DeptID	HireDate
1	102	Rahul	65000.00	1	2025-12-15
2	106	Vikas	95000.00	2	2025-10-01
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	110	Pooja	68000.00	4	2026-06-01

```

/*=====
5. INNER JOIN
=====*/

```

```

SELECT
    e.EmpID,
    e.EmpName,
    d.DeptName,
    e.Salary
FROM Employee e
INNER JOIN Department d
ON e.DeptID = d.DeptID;

```

Results		Messages		
	EmpID	EmpName	DeptName	Salary
1	101	Megha	IT	75000.00
2	102	Rahul	HR	65000.00
3	103	Amit	IT	90000.00
4	104	Priya	Finance	85000.00
5	105	Sneha	Sales	60000.00
6	106	Vikas	IT	95000.00
7	107	Neha	Marketing	70000.00
8	108	Kiran	HR	55000.00
9	109	Rohan	Finance	80000.00
10	110	Pooja	Sales	68000.00

/*=====

6. LEFT JOIN

=====*/

```
SELECT
    d.DeptID,
    d.DeptName,
    e.EmpName
FROM Department d
LEFT JOIN Employee e
ON d.DeptID = e.DeptID;
```

/*=====

7. GROUP BY WITH HAVING

DEPARTMENTS HAVING MORE THAN 1 EMPLOYEE

=====*/

```
SELECT
    DeptID,
    COUNT(*) AS TotalEmployees
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 1;
```

	DeptID	DeptName	EmpName
1	1	HR	Rahul
2	1	HR	Kiran
3	2	IT	Megha
4	2	IT	Amit
5	2	IT	Vikas
6	3	Finance	Priya
7	3	Finance	Rohan
8	4	Sales	Sneha
9	4	Sales	Pooja
10	5	Marketing	Neha
11	6	Admin	NULL
12	7	Operations	NULL
13	8	Support	NULL
14	9	Legal	NULL
15	10	Research	NULL

```

/*=====
7. GROUP BY WITH HAVING
DEPARTMENTS HAVING MORE THAN 1 EMPLOYEE
=====*/

```

```

SELECT
    DeptID,
    COUNT(*) AS TotalEmployees
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 1;

```

Results Messages		
	DeptID	TotalEmployees
1	1	2
2	2	3
3	3	2
4	4	2

```

/*=====
8. EMPLOYEES HIRED IN LAST 6 MONTHS
=====*/

```

```

SELECT *

```

```
FROM Employee
WHERE HireDate >= DATEADD(MONTH, -6, GETDATE());
```

Results		Messages			
	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	103	Amit	90000.00	2	2026-01-20
3	105	Sneha	60000.00	4	2026-03-05
4	107	Neha	70000.00	5	2026-04-12
5	108	Kiran	55000.00	1	2026-05-01
6	109	Rohan	80000.00	3	2026-01-15
7	110	Pooja	68000.00	4	2026-06-01

```
/*=====
9. INSERT DUPLICATE RECORDS FOR PRACTICE
=====*/
```

```
INSERT INTO Employee
VALUES
(111, 'Megha', 75000, 2, '2025-02-10'),
(112, 'Rahul', 65000, 1, '2024-12-15');
```

Messages	
(2 rows affected)	
Completion time: 2026-06-23T15:02:11.9122031+05:30	

	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01
11	111	Megha	75000.00	2	2025-02-10
12	112	Rahul	65000.00	1	2024-12-15

```

/*=====
10. FIND DUPLICATE RECORDS
=====*/

```

```

SELECT
    EmpName,
    Salary,
    COUNT(*) AS DuplicateCount
FROM Employee
GROUP BY EmpName, Salary
HAVING COUNT(*) > 1;

```

Results		Messages	
	EmpName	Salary	DuplicateCount
1	Rahul	65000.00	2
2	Megha	75000.00	2

```

/*=====
11. REMOVE DUPLICATE RECORDS
=====*/

```

```

WITH DuplicateRecords AS
(
    SELECT *,
        ROW_NUMBER() OVER
        (

```

```

PARTITION BY EmpName, Salary
ORDER BY EmpID
) AS RN
FROM Employee
)

```

```

DELETE
FROM DuplicateRecords
WHERE RN > 1;

```

Results		Messages			
	EmpID	EmpName	Salary	DeptID	HireDate
1	101	Megha	75000.00	2	2026-02-10
2	102	Rahul	65000.00	1	2025-12-15
3	103	Amit	90000.00	2	2026-01-20
4	104	Priya	85000.00	3	2025-11-10
5	105	Sneha	60000.00	4	2026-03-05
6	106	Vikas	95000.00	2	2025-10-01
7	107	Neha	70000.00	5	2026-04-12
8	108	Kiran	55000.00	1	2026-05-01
9	109	Rohan	80000.00	3	2026-01-15
10	110	Pooja	68000.00	4	2026-06-01